

De Morgan's theorems for any number of variables resemble in form the two-variable case and can be derived by successive substitutions similar to the method used in the above derivation.

These theorems can be generalized as follows:

$$(A+B+C+D+\dots+F)' = A'B'C'D'\dots F'$$

$$(ABCD\dots F)' = A'B'+C'+D'+\dots+F'$$

The generalized form of De Morgan's theorem states that the complement of a function is obtained by interchanging AND and OR operators and complementing each literal. $\hookrightarrow$ Imp

Example 2-3: Find the complement of the functions $F_1 = x'y'z' + x'y'z$ and $F_2 = x(y'z' + yz)$

Applying De Morgan's theorem as many times as necessary, the complements are obtained as follows:

$$F_1' = (x'y'z' + x'y'z)' = (x'y'z')' \cdot (x'y'z)'$$

$$F_2' = [x(y'z' + yz)]' = x' + (y'z' + yz)' = x' + (y'z')' \cdot (yz)'$$

A simpler procedure for deriving the complement of a function is to take the dual of the function and complement each literal. This method follows from the generalized De Morgan's theorem. Remember that the dual of a function is obtained from the interchange of AND and OR operators and 1's and 0's. $\hookrightarrow$ Imp

Example 2-4: Find the complement of the functions F_1 and F_2 of Example 2-3 by taking their duals and complementing each literal.

1. $F_1 = x'y'z' + x'y'z$
The dual of F_1 is $(x+y+z) \cdot (x+y+z)$
Complement each literal: $x' + y' + z' = F_1'$

2. $F_2 = x(y'z' + yz)$
The dual of F_2 is $x + (y+z)(y+z)$
Complement each literal: $x' + (y'+z') = F_2'$